

Exercise Sheet 2 - Solutions

Symbolic Logic

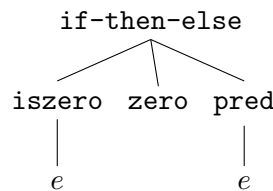
1. • The BNF for this language is defined as follows:

$$e ::= \text{zero} \mid \text{succ}(e) \mid \text{pred}(e) \mid \text{iszero}(e) \mid \text{if } e \text{ then } e \text{ else } e \mid \text{true} \mid \text{false}$$

- No expression is ambiguous using this grammar.
- The expression that given an expression e , checks whether e is zero, and if it is returns zero, else returns e 's predecessor, can be written as:

$$\text{if iszero}(e) \text{ then zero else pred}(e)$$

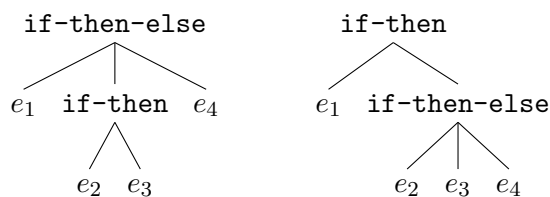
Its parse tree is



2. • The BNF for this extended language is defined as follows:

$$e ::= \text{zero} \mid \text{succ}(e) \mid \text{pred}(e) \mid \text{iszero}(e) \mid \text{if } e \text{ then } e \text{ else } e \mid \text{true} \mid \text{false} \mid \text{if } e \text{ then } e$$

- It now has ambiguities because expressions of the form $(\text{if } e_1 \text{ then if } e_2 \text{ then } e_3 \text{ else } e_4)$ could be $(\text{if } e_1 \text{ then } (\text{if } e_2 \text{ then } e_3) \text{ else } e_4)$ or $(\text{if } e_1 \text{ then } (\text{if } e_2 \text{ then } e_3 \text{ else } e_4))$.
- The parse trees corresponding to the two above expressions are:



3. • Our BNF is now:

$$\begin{aligned} e &::= \text{zero} \mid \text{succ}(e) \mid \text{pred}(e) \mid \text{iszero}(e) \mid \text{if } e \text{ then } e \text{ else } e \mid \text{true} \mid \text{false} \mid \text{if } e \text{ then } e \\ eq &::= e = e \end{aligned}$$

- An axiom schema capturing that “the expression that given an expression e , checks whether e is zero, and if it is returns zero, else returns e 's predecessor” is equal to “ e 's predecessor” is

$$\text{if iszero}(e) \text{ then zero else pred}(e) = \text{pred}(e)$$

This axiom schema makes use of one metavariable: e .

- Here are two instances of this axiom:
 - $\text{if iszero}(\text{zero}) \text{ then zero else pred}(\text{zero}) = \text{pred}(\text{zero})$
 - $\text{if iszero}(\text{succ}(\text{zero})) \text{ then zero else pred}(\text{succ}(\text{zero})) = \text{pred}(\text{succ}(\text{zero}))$